

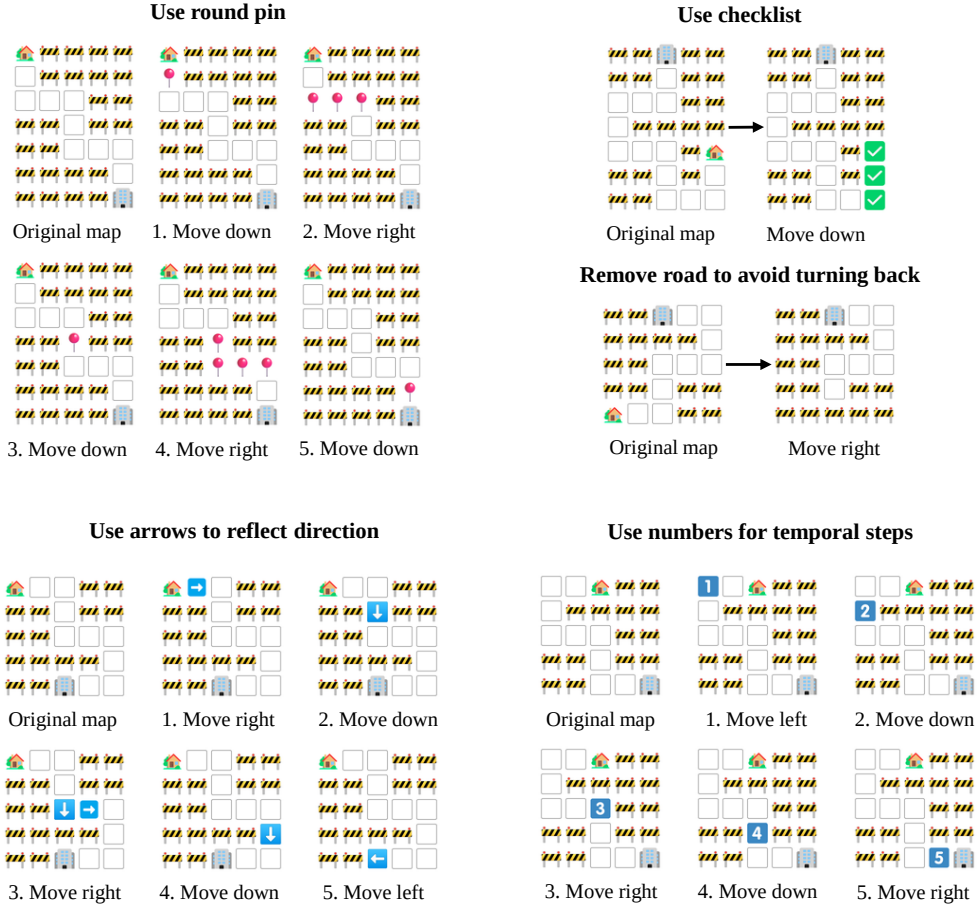


Figure 10: *Mental images* for state tracking in visual navigation tasks. We found diverse visualization formats in LLMs to track the navigation progress, including marking the path, marking the current location, marking directional steps, and marking temporal steps. More details are available in Appendix E.1

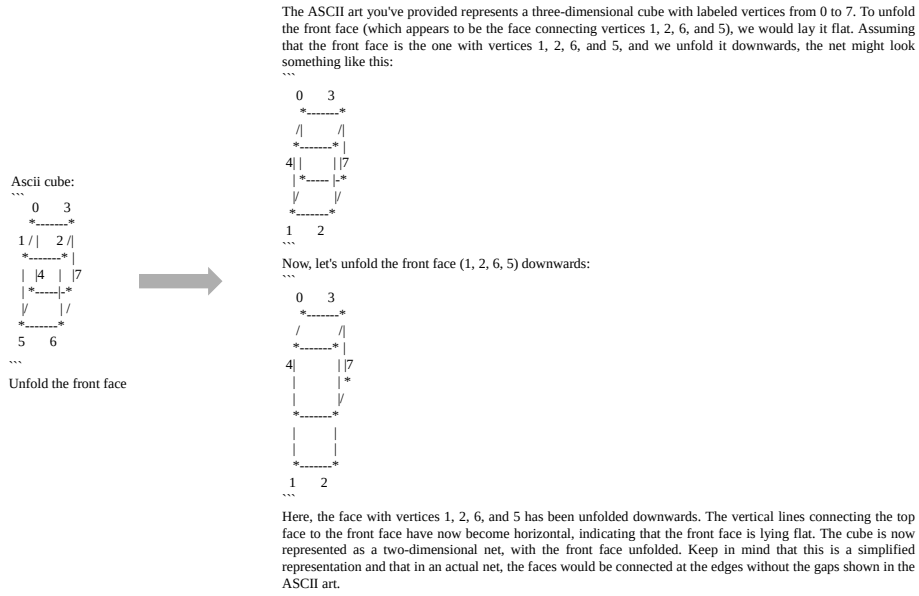


Figure 11: Representation for 3D semantics and qualitative results of GPT-4 manipulating the *mental image* correctly.

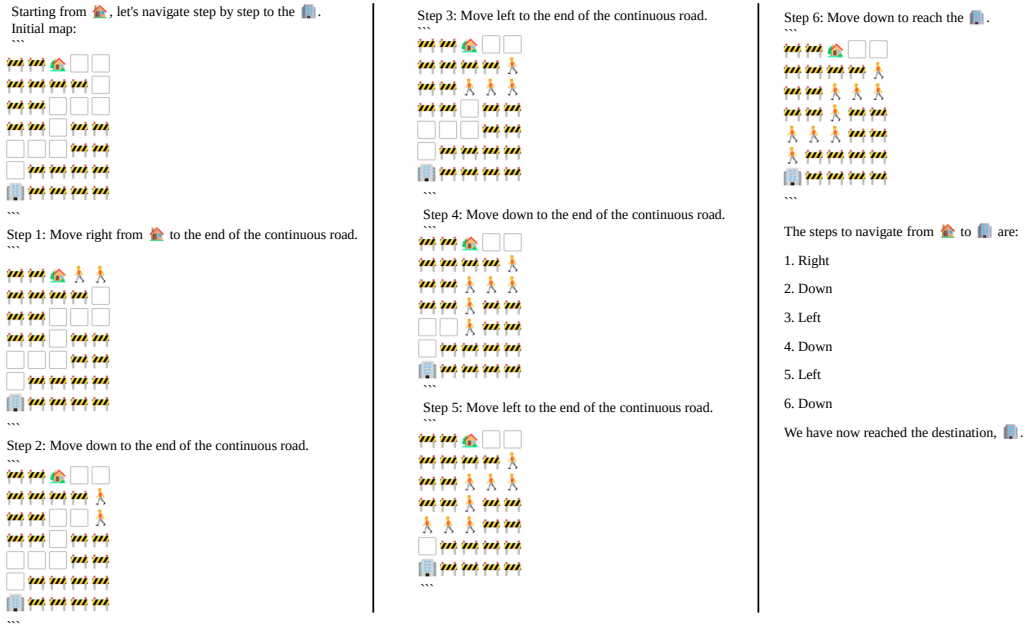


Figure 12: Responses of Route Planning

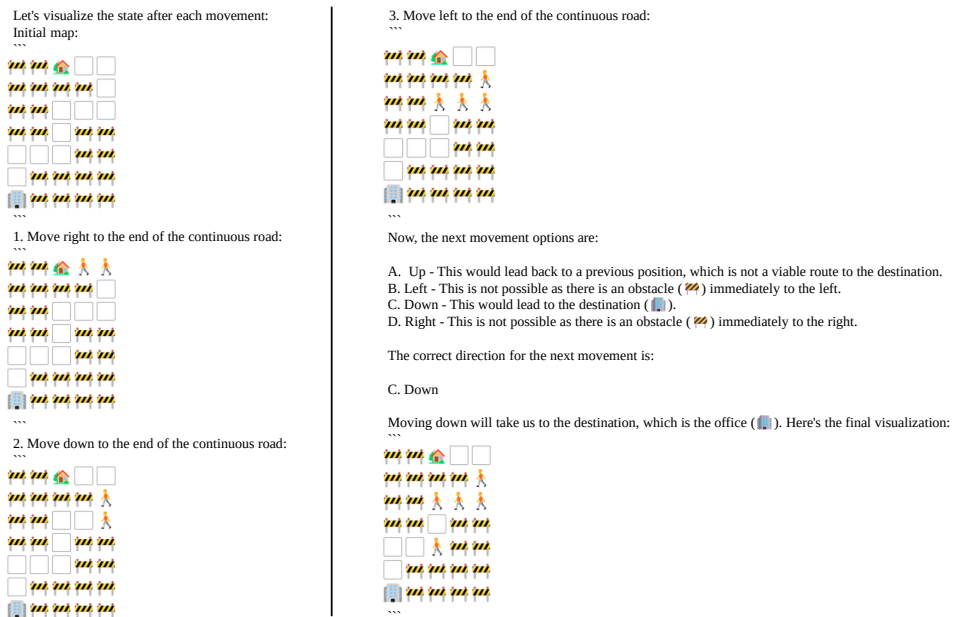


Figure 13: Response of Next Step Prediction